FCK-U SYSTEM

Frame Cab Kinetic Utilization

Open Source Prior Art Technical Disclosure

ORIGINATOR: [Your Name]

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STATUS: Intentionally Released — No Rights Reserved

This document constitutes a public disclosure of the inventive concepts described herein. Publication establishes prior art. No entity may patent these concepts. All manufacturers, engineers, and researchers are free to implement, modify, and build upon this work.

1. FIELD OF THE INVENTION

This disclosure relates to a novel vehicle architecture combining an actively decoupled passenger cabin, multi-stage adaptive vibration isolation, omnidirectional crash-displacement safety systems, and a federated dual-ECU control architecture enabling the passenger cabin to function as an autonomous survival pod upon separation from the vehicle frame.

Target application: Heavy-duty electric vehicle platforms with steer-by-wire and drive-by-wire architectures, including but not limited to the Tesla Cybertruck platform and similar body-on-frame or unibody EV platforms.

2. BACKGROUND AND PRIOR ART

Conventional vehicle chassis design couples the passenger cabin rigidly to the frame. While modern vehicles incorporate passive and semi-active suspension systems, the cabin itself remains a fixed structural element. This creates three persistent problems:

- Vibration transmission: Road-surface irregularities, drivetrain harmonics, and chassis flex are transmitted directly to the occupant compartment with only seat-level attenuation.
- Fixed collision geometry: In a frontal impact, the occupant cabin decelerates at the same rate as the frame. The crumple zone provides the only buffer between impact energy and the passenger.
- Battery-proximate fire risk: In battery electric vehicles, the traction battery pack is integrated into the structural floor, placing occupants in direct proximity to the highest thermal-runaway risk component.

Existing semi-active suspension systems (Bose, MagneRide, Audi eROT) address vibration at the wheel/frame interface but do not decouple the cabin. Active safety systems (Tesla Autopilot collision avoidance, pre-tensioning seatbelts) prepare occupants for impacts but do not physically alter cabin geometry during a crash. No prior art exists for a unified system that decouples the cabin from both road vibration and crash energy while simultaneously separating it from the battery pack.

3. SUMMARY OF THE INVENTION

The FCK-U System (Frame Cab Kinetic Utilization) is a multi-layer vehicle architecture comprising:

- A tri-stage cooperative adaptive suspension isolating the cabin from road inputs across all frequency ranges
- A gimbaled Center-of-Gravity swivel yoke providing the mechanical pivot for cabin independence
- An active kinetic displacement system enabling omnidirectional cabin separation during impact
- A composite fire-barrier skid plate forming the structural base of the separated cabin
- A federated dual-ECU architecture where a cabin-resident secondary ECU assumes autonomous control post-separation
- A dual-loop regenerative energy harvesting system powering a dedicated cabin survival battery

4. DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

4.1 Tri-Stage Cooperative Adaptive Suspension (E-CESA / Yoke / Pivot)

The suspension system is divided into three cooperative stages, each responsible for a distinct frequency domain. All three stages communicate on a shared real-time CAN bus and are coordinated by a Conflict Resolver running at a minimum of 1,000 Hz.

Stage 1: E-CESA (ECU-Controlled Electric Shock Absorbers)

Location: Frame mounting points, all four corners.

Frequency domain: Low-frequency, high-energy events (0.5 – 5 Hz). Potholes, speed bumps, body roll, braking dive.

Operation: Variable-rate electromagnetic dampers controlled by the main ECU via sensor fusion inputs including IMU, wheel speed sensors (WSS), brake pressure transducers, and steering angle sensors. The E-CESA pre-charges to the predicted required damping rate before the event occurs, eliminating reactive lag.

Energy function: Primary regenerative loop. Each E-CESA strut contains electromagnetic induction coils. At typical road inputs, each strut generates an estimated 100W – 500W. Output is rectified and fed to the main traction battery via DC-DC converter, providing measurable range extension.

Stage 2: Yoke Soft Struts (Dual-Angled Linear Generators)

Location: Attached between the CG swivel pivot assembly and the cabin floor structure.

Frequency domain: High-frequency, low-energy events (15 – 150 Hz). Gravel surface chatter, road texture hum, minor wheel imbalance, drivetrain vibration.

Operation: Dual angled struts arranged in a V-configuration act as a mechanical differential for oscillation, providing opposing force vectors that cancel lateral micro-oscillations. Strut compliance is electrically variable.

Energy function: Secondary regenerative loop. Continuous high-frequency motion drives linear generator coils. Output charges the isolated Cabin Survival Battery (12V/48V LFP chemistry). This secondary battery is dedicated exclusively to cabin safety systems and is physically decoupled from the main traction battery.

Stage 3: Active Pivot Bearing (CG Swivel Yoke)

Location: Single central mounting point aligned with the cabin's calculated Center of Gravity.

Frequency domain: Rotational and torsional inputs across all frequencies.

Operation: A gimbaled pivot bearing with electronically variable resistance on all rotational axes. In normal operation, the pivot allows the chassis to roll and pitch while maintaining the cabin floor in a near-horizontal orientation (Skyhook model). In crash mode, the pivot resistance is zeroed to allow free omnidirectional displacement.

Cooperative Conflict Resolver Logic

A dedicated software layer assigns a dominance percentage to each stage based on real-time frequency analysis of incoming road data:

Input Frequency	E-CESA Dominance	Yoke Dominance	Pivot Activity
0.5 – 5 Hz (macro events)	80%	20%	Stabilizing
5 – 15 Hz (transition zone)	50%	50%	Adaptive
15 – 150 Hz (micro events)	20%	80%	Passive
Crash event (>5G detected)	0% — Actuator mode	0% — Locked	Free — Full release

Stages are never permitted to operate at conflicting settings simultaneously. The resolver enforces alternating dominance to prevent inter-stage resonance amplification.

4.2 Center of Gravity Swivel Pivot — Mathematical Model

The pivot must be located at the true dynamic CG of the loaded cabin to minimize the restoring moment required to maintain level orientation. The CG location is not fixed and must be calculated continuously.

Static CG height (z-axis):

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    z_cg = (m_structure × z_structure + m_occupants × z_occupants +
m_battery × z_battery) / m_total
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The pivot bearing assembly uses a motorized vertical adjustment mechanism with ±80mm of travel to track CG shifts from variable occupant load (0–5 passengers) and cargo. Adjustment rate: 10mm/sec continuous, 80mm/sec emergency.

Lateral CG compensation (x/y axes): The Yoke strut compliance is differentially adjusted in real-time to compensate for lateral CG shifts, applying a bias force that keeps the cabin floor level without requiring physical pivot relocation.

4.3 FCK-U Active Kinetic Displacement — Omnidirectional Breakaway System

This system replaces the prior concept of a linear rail-and-sled with a four-point omnidirectional pyrotechnic mount system. This provides significantly greater flexibility in impact response direction.

Mounting Architecture

The cabin is attached to the skid plate base via four pyrotechnic mounting points, one at each corner. Each mount consists of:

- A structural shear bolt rated to 200 kN static load
- A co-located pyrotechnic charge with independently addressable firing circuit
- A guide cone that maintains cabin orientation during the first 50–80mm of displacement

Impact Vector Analysis and Firing Sequence

The Cabin ECU continuously monitors impact vector data from the forward LIDAR/RADAR array. Upon confirmation of unavoidable collision, the firing sequence is determined by impact azimuth:

Impact Direction	Firing Sequence	Cabin Displacement Direction	Priority
Direct frontal (0°)	All 4 simultaneous	Rearward	Maximum separation
Left offset (<45°)	Left mounts first, then right	Rearward-right	Angular escape
Right offset (>315°)	Right mounts first, then left	Rearward-left	Angular escape

Impact Direction	Firing Sequence	Cabin Displacement Direction	Priority
Rear impact	All 4 simultaneous	Forward ejection	Battery separation priority
Rollover detected	All 4 simultaneous	Drop to skid plate	Ground contact stabilization

All four charges are capable of firing within 20ms of the trigger signal. Selective sequencing introduces a 5–10ms delay on secondary mounts to produce the desired angular displacement vector.

Force Requirements

For a cabin mass of approximately 600 kg and a target displacement of 0.3m (12 inches):

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Required actuator force = m × a_crash = 600 kg × 245 m/s² =
147,000 N (33,000 lbf)
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This force requirement is compatible with automotive-grade pyrotechnic actuators (similar in principle to seatbelt pretensioners, scaled appropriately). Electric linear actuators are insufficient for this application due to force and response time requirements.

4.4 Composite Fire-Barrier Skid Plate

The skid plate serves as the structural floor of the separated cabin, replacing the load-bearing function previously provided by the battery sled/frame. It is a permanent structural component of the cabin assembly.

Material Specification

- Outer shell: Ceramic-coated carbon fiber composite, minimum 8mm thickness
- Core: Closed-cell polymer foam (polyisocyanurate or equivalent), minimum 40mm thickness
- Inner liner: 3mm aluminum honeycomb for impact distribution
- Fire resistance: Outer ceramic coating rated to minimum 1100°C continuous, 1600°C short-term
- Buoyancy: Closed-cell foam core provides positive buoyancy for the full cabin assembly when sealed

Functional Roles

- Structural: Primary load-bearing floor of the separated cabin, replacing battery sled structural contribution
- Thermal barrier: Prevents thermal runaway fire propagation from battery pack below to occupant compartment
- Sliding surface: Low-friction ceramic outer surface allows the cabin to slide across asphalt, gravel, grass, and wet surfaces with predictable deceleration characteristics
- Flotation: Sealed foam core provides emergency buoyancy in water-immersion scenarios (bridge collapse, flood)

4.5 Battery-Cabin Physical Separation

In current EV architecture (notably the Cybertruck platform), the traction battery is a structural element integrated into the vehicle floor. The FCK-U system requires a redesign of this interface to allow clean physical separation.

Interface Design

The battery sled and cabin floor meet at a defined Separation Interface Plane (SIP). This interface contains:

- High-voltage pyro disconnect (fires first in sequence — electrical separation precedes mechanical)
- Low-voltage/data quick-disconnect (blind-mate connector, releases under 50N pull force post-pyro)
- Cooling loop breakaway valves (spring-loaded, seal automatically on separation to prevent coolant spray)
- Mechanical mounts (pyrotechnic, fire last in sequence)

Post-Separation Battery Behavior

After cabin separation, the battery sled remains with the frame/chassis. The main ECU, also frame-resident, executes the following shutdown sequence:

- Immediately: Contactor opens, HV bus de-energized
- T+100ms: Cooling pump maximum flow to battery pack
- T+500ms: Thermal runaway monitoring continues via frame-resident sensors
- T+1s: Emergency beacon activates on frame with GPS coordinates, battery state, and thermal status

4.6 Federated Dual-ECU Architecture

System Overview

The vehicle operates two ECUs in a federated relationship. Under normal conditions, the Main ECU holds primary authority. The Cabin ECU operates in warm standby, mirroring relevant state data.

System / Function	Normal Mode Authority	Post-Separation Authority
Traction motors	Main ECU	Disabled (no connection)
Main battery management	Main ECU	Main ECU (frame-resident)
Steer-by-wire	Main ECU (primary)	Cabin ECU (full authority)
Brake-by-wire	Main ECU (primary)	Cabin ECU (full authority)
Airbag deployment	Both ECUs in parallel	Cabin ECU (full authority)
Door e-latches	Main ECU	Cabin ECU
Windows	Main ECU	Cabin ECU
Emergency beacon / GPS SOS	Main ECU	Both independent beacons
Cabin survival battery	Charging from main	Sole power source
60GHz wireless link	Standby	Primary control link

Cabin ECU Power Budget

The Cabin ECU and all cabin-only systems are powered by the Survival Battery. Estimated post-separation load:

System	Power Draw	Duration Required
Steer-by-wire actuators	~200W peak, ~80W continuous	Until vehicle stops (~60s)
Brake-by-wire	~150W peak	Until vehicle stops
Emergency lighting	~30W	30 minutes
60GHz wireless link	~15W	Until vehicle stops + 5 min
GPS / Emergency beacon	~10W	2 hours minimum
Airbag system (post-fire)	~5W standby	30 minutes
Total peak	~490W	—

A 2 kWh LFP Survival Battery (approximately the physical size of a large motorcycle battery) is sufficient for all post-separation functions with significant margin. LFP chemistry is selected for minimal thermal runaway risk.

4.7 W-FCK-U Wireless Redundancy System

The 60GHz millimeter-wave wireless link provides the control pathway for steer-by-wire and brake-by-wire in the event that the physical umbilical is severed during separation.

Latency Analysis at 80mph (36 m/s)

Link Layer	Latency	Notes
60GHz RF propagation (cabin to frame)	~1ms	Line-of-sight, sub-meter distance
Protocol stack overhead	~5–8ms	IEEE 802.11ad/WiGig protocol
Steer actuator response time	~15–20ms	Hydraulic-assist or electric rack
Total round-trip latency	~21–29ms	Well within 50ms human perception threshold

Antenna Array Requirement

60GHz signals are susceptible to blockage by metallic deformation and water. To ensure link availability during and after crash deformation, a minimum of three antenna nodes are required on both cabin and frame sides, positioned to ensure at least one antenna pair maintains line-of-sight regardless of cabin displacement angle.

- Cabin antenna positions: Front-center, rear-left, rear-right
- Frame antenna positions: Front-left, front-right, rear-center
- Link management: Automatic antenna diversity switching, <1ms switchover time

4.8 500ms FCK-U Safety Protocol — Full Sequence

Time	Event	System Responsible	Action
T-500ms	Threat confirmed	LIDAR/RADAR + Main ECU	E-CESA enters ultra-stiff mode. Cabin ECU enters full-authority standby.
T-250ms	Pre-impact prepare	Cabin ECU	60GHz link enters Active-Active state. Survival battery disconnects from charging loop.
T-100ms	Imminent impact	Cabin ECU	Emergency lighting activates. Airbag system arms. Seatbelt pretensioners fire.
T-0ms	Impact	Both ECUs	Pyro sequence initiates. HV disconnect fires first.
T+5ms	HV isolated	Main ECU	High-voltage bus de-energized. Cooling pump maximum flow.

Time	Event	System Responsible	Action
T+10ms	Data disconnect	Both ECUs	Umbilical data link severs. 60GHz assumes full control authority.
T+20ms	Mechanical separation	Cabin ECU	Pyrotechnic mounts fire in calculated sequence. Cabin displaces on skid plate.
T+50ms	Cabin free	Cabin ECU	Cabin sliding on skid plate. Steer/brake-by-wire active via wireless.
T+500ms	Vehicle at rest	Both ECUs (independent)	Cabin ECU: pop doors, lower windows, activate SOS beacon. Main ECU: frame beacon + thermal monitoring.

4.9 Self-Learning ECU and Predictive Suspension Logic

The primary ECU incorporates a machine learning model for route-adaptive suspension optimization. This model operates analogously to an automotive fuel-trim system, building a frequency-domain map of road surfaces encountered on repeated routes.

- Route memory: GPS-tagged road surface frequency profiles stored and updated after each traversal
- Pre-event adjustment: Suspension settings adjusted 200–500ms before known road events (specific potholes, rough sections, driveway entries) based on GPS position match
- Occupant modeling: Learns typical occupant load profile to refine CG calculations over time
- Curve anticipation: Differential wheel speed sensor data used to detect curve entry and apply predictive cabin lean angle to cancel centrifugal force on occupants

5. CLAIMS (Non-Binding — Prior Art Disclosure)

The following claims define the scope of the inventive concepts being placed into the public domain. No patent protection is sought. These claims are stated to define the boundaries of prior art established by this disclosure.

1. A vehicle architecture comprising a passenger cabin mechanically decoupled from the vehicle frame via a gimbaled swivel pivot located at the dynamic Center of Gravity of the cabin assembly.

2. A tri-stage cooperative adaptive suspension system wherein three vibration isolation stages share a real-time control bus and operate under a Conflict Resolver algorithm that enforces alternating stage dominance to prevent inter-stage resonance.
3. An omnidirectional pyrotechnic cabin displacement system using four independently addressable corner mounts that fires in calculated sequence based on impact azimuth to direct cabin movement away from the collision vector.
4. A composite skid plate forming the structural floor of a separated passenger cabin, providing simultaneous fire barrier, sliding surface, and positive buoyancy functions.
5. A physical separation interface between the passenger cabin and the traction battery pack, wherein the cabin separates from and moves away from the battery pack upon impact, and wherein the separation sequence prioritizes high-voltage electrical isolation before mechanical disconnection.
6. A federated dual-ECU architecture wherein a Cabin-resident secondary ECU operates in warm standby during normal operation and assumes full authority over steer-by-wire, brake-by-wire, and occupant safety systems upon cabin separation from the vehicle frame.
7. A dual-loop regenerative energy harvesting system wherein primary-loop electromagnetic struts charge the main traction battery and secondary-loop linear generator struts charge an isolated Cabin Survival Battery dedicated to post-separation safety functions.
8. A 60GHz millimeter-wave wireless control link with minimum three-node antenna diversity on each side of the cabin-frame separation plane, providing steer-by-wire and brake-by-wire control continuity through and after cabin separation.

6. OPEN SOURCE LICENSE AND PUBLIC DOMAIN DECLARATION

This disclosure is made with the explicit intent of placing all described inventive concepts into the public domain. The originator expressly waives all patent rights to these concepts. Any person, company, or institution is free to implement, manufacture, license, sublicense, or build upon these concepts without restriction, royalty, or attribution requirement.

The publication of this document constitutes prior art under 35 U.S.C. § 102 (United States) and equivalent statutes internationally. No entity may obtain valid patent protection for the concepts described herein after the date of this publication.

The originator provides this disclosure in the interest of public safety and encourages all vehicle manufacturers to evaluate and implement these concepts as rapidly as possible.

DISCLAIMER: This is a conceptual engineering disclosure, not a certified engineering specification. Manufacturers implementing these concepts are solely responsible for all engineering validation, safety certification, regulatory compliance, and testing required for production deployment.

7. ORIGINATOR INFORMATION & TIMESTAMP

Field	Information
Originator	[Your Full Name]
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Document Version	1.0 — Initial Public Disclosure
Platform	X (Twitter) — @[YourHandle] | #OpenSourceSafety #FCKU
Contact	[Optional — email or website]
Collaborating AI Systems	Gemini (concept development), Claude (engineering analysis and refinement)

— END OF DISCLOSURE —

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